

Lecture 3

Describing and Summarizing Data

Part 2.

Measures of Location and Dispersion

| Location and Dispersion

Descriptive statistics often summarizes a variable with only a few numbers.

Two key questions are:

- ❑ **Where is the data located?** → measures of location: mean, median, mode
- ❑ **How spread out is the data?** → measures of dispersion: range, variance, standard deviation, interquartile range

Together, location and dispersion provide a **compact representation of the data**.

However, they do not describe the full shape of the distribution. Only for special distributions, such as the *normal distribution*, can a small number of summary statistics describe most of what we need to know.

Measures of Location

Arithmetic mean (also called average):

- Sum of all numbers divided by size of dataset.
- Applies only to numeric data (discrete or continuous,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Median:

- Middle value when data arranged in order
- Applies to ordinal and numeric data)

$$\tilde{x} = \begin{cases} x_{(\frac{n+1}{2})}, & \text{if } n \text{ is odd,} \\ \frac{x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}}{2}, & \text{if } n \text{ is even.} \end{cases}$$

Mode:

- Most frequently occurring value
- Can be used with qualitative data, and numeric, but for continuous data, it must be applied to the distribution (or frequency table)

$$\text{mode} = \arg \max_y f(y)$$

Raw numeric Data – an example

Dataset #1 (N=11):

1,2,4,5,5,5,6,6,8,9,10

The **mean** is: $\text{Sum}/N = 61/11 = 5,545$

The **median** is the middle one, that is, **5**

1,2,4,5,5,5,6,6,8,9,10



The **mode** is the most frequent value, also **5**

Dataset #2 (N=12)

1,2,4,5,5,5,6,6,8,9,10, 35

identical to dataset 1 except one number)

The **mean** becomes $96/12 = 8$

The **median** is between the numbers 5 and 6, so we *decide* **5.5**

1,2,4,5,5,5,6,6,8,9,10,35



The mode remains at **5**

- ❑ In roughly symmetric datasets, the mean, median, and mode are often similar.
- ❑ However, the mean is not robust to extreme values: one unusually large or small observation can change it substantially.
- ❑ The median is usually more stable in the presence of outliers.

Mean from Tabulated Data

When data are tabulated, each value (or class) is associated with a **frequency**.

The arithmetic mean becomes a **weighted average**, where each value contributes according to how often it occurs.

Items bought	Frequency	$x_i f_i$
1	6	6
2	14	28
3	21	63
4	12	48
5	7	35
Totals	60	180

Formula

$$\bar{x} = \frac{\sum_{i=1}^k x_i f_i}{\sum_{i=1}^k f_i} = \frac{\sum_{i=1}^k x_i f_i}{n}$$

Notation

x_i :value (or class midpoint for grouped data)

f_i :frequency of x_i

k :number of distinct values (or classes)

$n = \sum_{i=1}^k f_i$:total number of observations

$$\bar{x} = \frac{\sum_{i=1}^k x_i f_i}{\sum_{i=1}^k f_i} = \frac{6 + 28 + 63 + 48 + 35}{60} = \frac{180}{60} = 3.$$

Percentiles and Quartiles

Percentiles describe the **position** of a value within an ordered dataset.

The **p-th percentile** is a value such that approximately **(p%)** of the observations are at or below it.

Important special cases

- ❑ **(Q_1)** = first quartile = **25th percentile**
- ❑ **(Q_2)** = second quartile = **50th percentile** = **median**
- ❑ **(Q_3)** = third quartile = **75th percentile**

Percentiles are useful for describing the **distribution** of the data, not only its center.

To compute percentiles,:

- ❑ we first sort the data in increasing order,
- ❑ then locate the position corresponding to the desired percentage.

```
import numpy as np

x = [4, 7, 9, 10, 15, 18, 21]

p25 = np.percentile(x, 25) # 25th percentile = Q1
p50 = np.percentile(x, 50) # 50th percentile = median
p75 = np.percentile(x, 75) # 75th percentile = Q3

print(p25, p50, p75)
```

in python use the numpy function `np.percentile`

Measures of Dispersion

A measure of dispersion can be seen as a measure of **RISK**. If I take a randomly selected data point from my dataset (or in the future), how far can I get from the typical/average value?

Let us suppose the arithmetic mean is known and denoted by $\bar{x}$

Variance measures the average squared distance of the observations from the mean.

$$\text{Variance} = \frac{1}{N} \sum_i (x_i - \bar{x})^2$$

Standard deviation Typical distance of the observations from the mean, in the original units

$$\text{Standard Deviation or } \sigma = \sqrt{\frac{1}{N} \sum_i (x_i - \bar{x})^2} = \sqrt{\text{Variance}}$$

Range Distance between the largest and the smallest value in the dataset.

$$\text{Range} = \max(X) - \min(X)$$

Coefficient of variation Dispersion relative to the mean,

$$\text{Coefficient of Variation} = \frac{\sigma}{\bar{x}}$$

Subtle point in the definitions

For statistical and profound reasons, there are **2 competing definitions** of variance and standard deviation for samples of finite size.

a) {

$$\text{Variance} = \frac{1}{N} \sum_i (x_i - \bar{x})^2$$
$$\sigma = \sqrt{\frac{1}{N} \sum_i (x_i - \bar{x})^2}$$

The difference is always small specially for large datasets ($N > 30$)

However, some python libraries/functions implement **a)** and some other implements **b).**

Usually this is not an issue, but do not be surprised if you see slightly different results depending on the function that you are using !

b) {

$$\text{Variance} = \frac{1}{N-1} \sum_i (x_i - \bar{x})^2$$
$$\sigma = \sqrt{\frac{1}{N-1} \sum_i (x_i - \bar{x})^2}$$

Example:

`numpy.std` and `numpy.var` use a)
but `numpy.cov` uses b)

```
print(np.cov(x,x)[0,0])  
print(np.var(x))
```

```
1.016063103836798  
1.0059024727984303
```